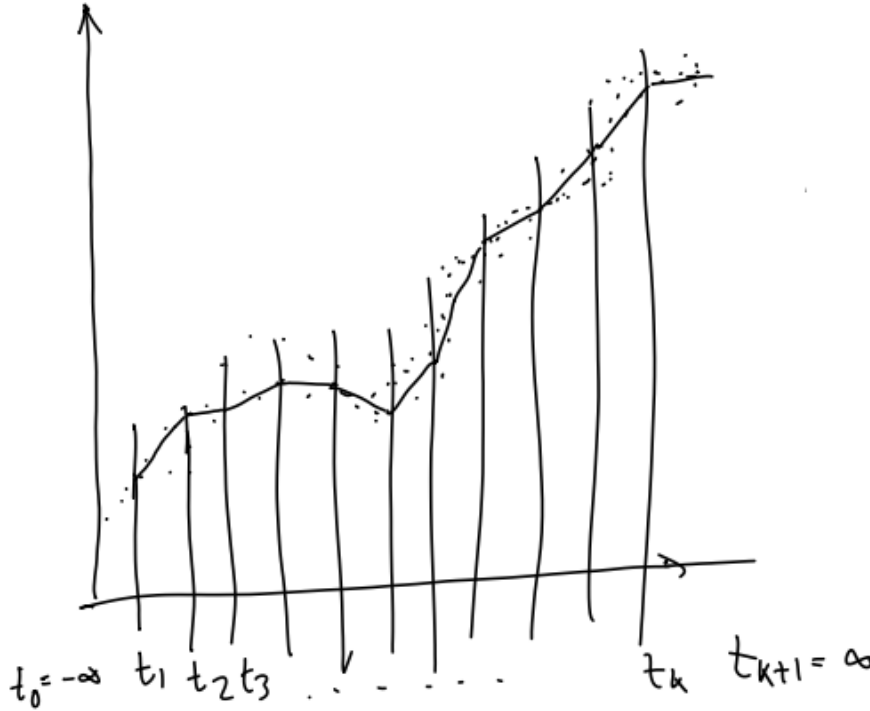


1 Linear Splines and Truncated Power Basis

The collection of functions



$$\{1, x, (x - t_1)_+, (x - t_2)_+, \dots, (x - t_k)_+\}$$

forms a basis for the class of functions we are interested in.

Now any function f which is linear in the interval (t_{i-1}, t_i) for all i and continuous at the t_i 's can be written as

$$f(x) = \alpha_0 + \alpha_1 x + \alpha_2 (x - t_1)_+ + \alpha_3 (x - t_2)_+ + \dots + \alpha_{k+1} (x - t_k)_+.$$

In L.S. regression, we minimize

$$\sum (Y_i - f(x_i))^2.$$

So, here we would like to minimize

$$\sum [Y_i - \alpha_0 - \alpha_1 x_i - \alpha_2 (x_i - t_1)_+ - \dots - \alpha_{k+1} (x_i - t_k)_+]^2.$$

In matrix form

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_1 & (x_1 - t_1)_+ & \dots & (x_1 - t_k)_+ \\ 1 & x_2 & (x_2 - t_1)_+ & \dots & (x_2 - t_k)_+ \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & (x_n - t_1)_+ & \dots & (x_n - t_k)_+ \end{bmatrix}, \quad \beta = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{k+1} \end{bmatrix}.$$

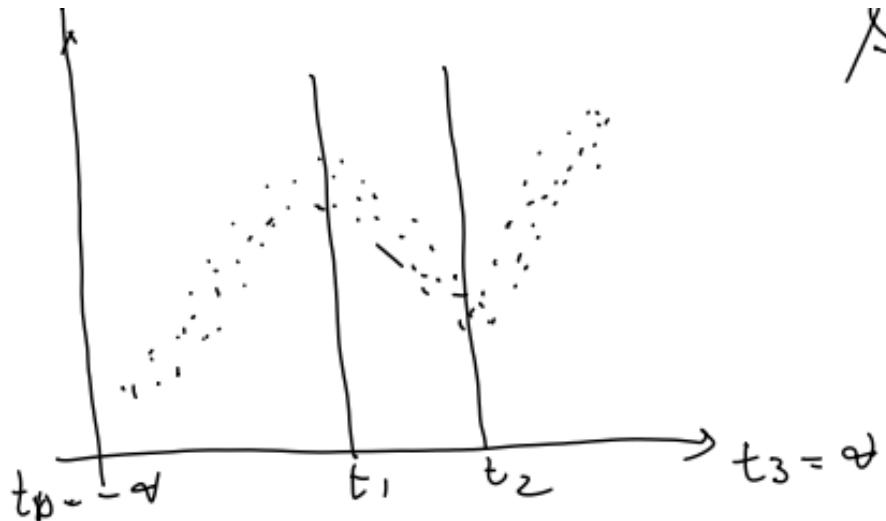
The ordinary least-squares estimator is

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

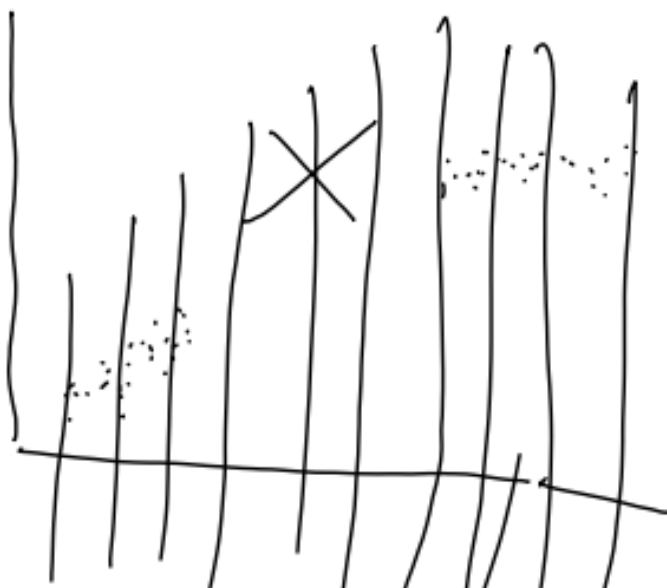
Questions about knots

- How many knots (what is the value of k)?
- Where should we place these knots?

Sometimes the scatter plot of the data set gives you an idea.

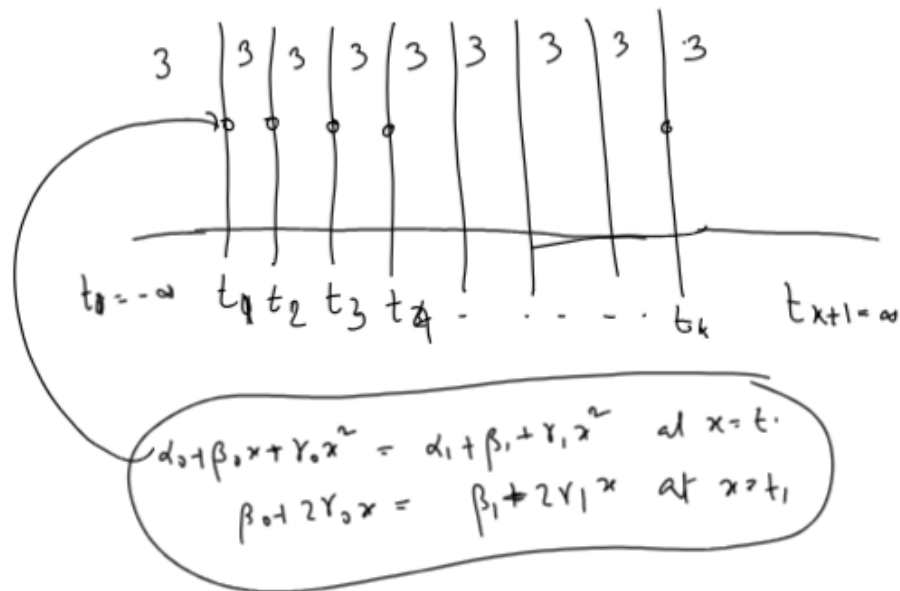


- We can start with sufficiently large number of knots and then use the backward deletion method to get a nested sequence of models. We use appropriate criterion (e.g. adjusted R^2) to select one of these models.
- Taking equi-spaced knots is not always a good option. It is better to place the knots based on order statistics.



Degrees of freedom counting:

$$3(k+1) - 2k = k + 3.$$



The collection of functions

$$\{1, x, x^2, \dots\}$$

2 Quadratic & Cubic Splines; Multivariate Extension

Similarly, for cubic splines we can consider knots at $t_1, t_2, \dots, t_k$. We can consider the basis

$$\{1, x, x^2, x^3, (x - t_1)_+^3, (x - t_2)_+^3, \dots, (x - t_k)_+^3\}.$$

This forms a basis for the quadratic/cubic splines.

This basis is called **power spline basis** or **truncated power spline basis**.

Indeed one can use other basis functions (e.g. B-spline basis) for regression.

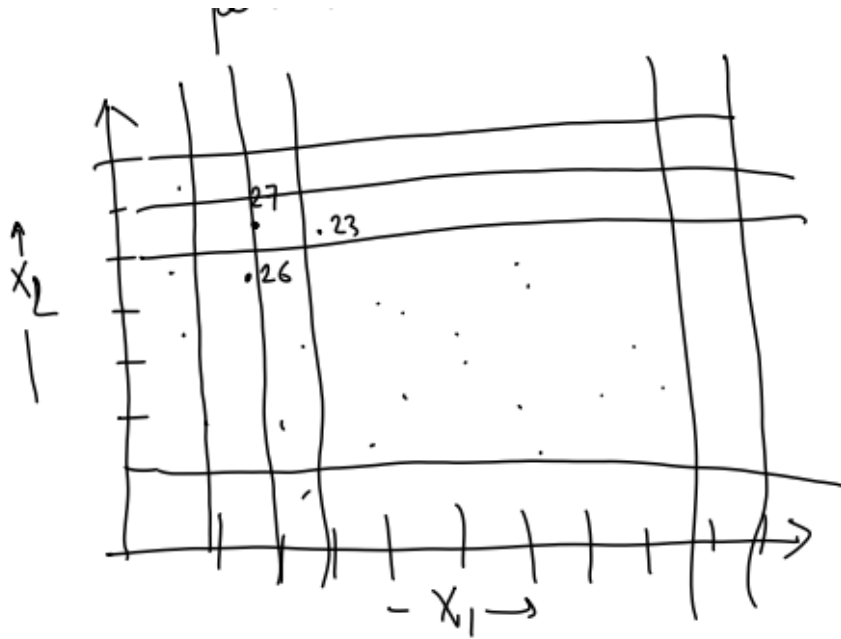
More than one predictor

Now consider the case when there are more than two (or more) predictor variables.

In two dimensions we can consider the basis

$$1, x_1, (x_1 - t_{11})_+, \dots, (x_1 - t_{1k})_+, \\ x_2, (x_2 - t_{21})_+, \dots, (x_2 - t_{2k})_+$$

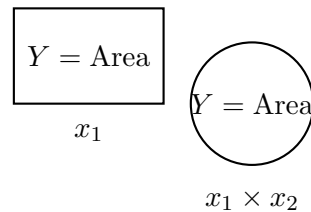
and go for linear regression as before.



If we do that then basically we are assuming that f is of the form

$$\begin{aligned} f(x_1, x_2) &= \alpha_0 + \alpha_1 x_1 + \alpha_2 (x_1 - t_{11})_+ + \cdots + \alpha_{k+1} (x_1 - t_{1k})_+ \\ &\quad + \beta_1 x_2 + \beta_2 (x_2 - t_{21})_+ + \cdots + \beta_{k+1} (x_2 - t_{2k})_+ \\ &= f_1(x_1) + f_2(x_2). \end{aligned}$$

So, here we are assuming an **additive model**.



However, this method won't work if we have functions like $f(x_1, x_2) = x_1 x_2$ (non additive models):

$$\begin{aligned} f(x_1, x_2) &\neq f_1(x_1) + f_2(x_2), \\ x_1 x_2 &\neq f_1(x_1) + f_2(x_2). \end{aligned}$$

But we can rewrite the product as a sum of functions of linear projections:

$$\begin{aligned} x_1 x_2 &= \left(\frac{x_1 + x_2}{2} \right)^2 - \left(\frac{x_1 - x_2}{2} \right)^2 \\ &= f_1(\alpha_1^\top x) + f_2(\alpha_2^\top x), \end{aligned}$$

where

$$\begin{aligned} \alpha_1 &= \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \quad \alpha_1^\top x = \frac{x_1 + x_2}{\sqrt{2}}, \quad f_1(t) = \frac{t^2}{2}, \\ \alpha_2 &= \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right), \quad \alpha_2^\top x = \frac{x_1 - x_2}{\sqrt{2}}, \quad f_2(t) = -\frac{t^2}{2}. \end{aligned}$$

Additive model:

$$f(x_1, x_2) = f_1(x_1) + f_2(x_2).$$

In general

$$f(x_1, x_2) = \sum_{k=1}^M f_k(\alpha_k^\top x).$$

3 Projection Pursuit Regression & Single-Layer Perceptron

So instead of considering an additive model, one can assume that the underlying regression function f is of the form

$$f(x_1, \dots, x_p) = \sum_{k=1}^M f_k(\alpha_k^\top x), \quad \alpha_k \in \mathbb{R}^p, \quad x = (x_1, \dots, x_p), \quad M \in \mathbb{N}.$$

This type of model is known as the **projection pursuit model**.

- How to estimate the α_k 's, f_k 's and M ?

Single-layer perceptron

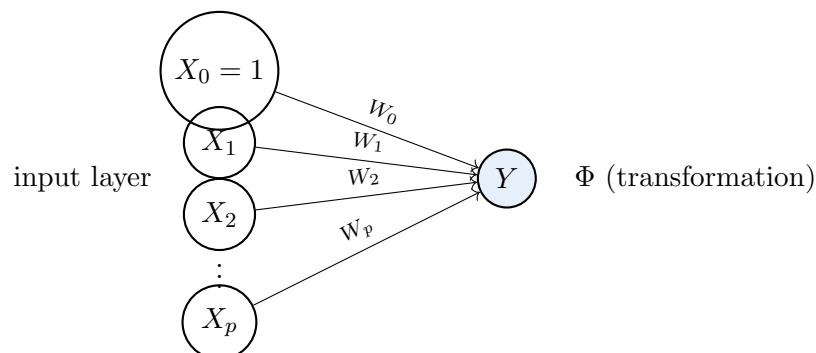
Input–output relationship:

$$Y = \Phi\left(W_0 + \sum_{i=1}^p W_i X_i\right).$$

In a single-layer perceptron model, we have the input–output relationship given by

$$Y = \Phi\left(W_0 + \sum W_i X_i\right).$$

This may be viewed as a generalized linear model.



For $\Phi(t) = t$ we get back the linear model.

Often

$$\Phi(t) = \frac{1}{1 + e^{-t}}$$

is used in the case of classification. We called it the **logistic function** / **sigmoid function**.

Classification rule:

$$\begin{aligned} Y &\leq C \quad \text{or} \quad Y > C, \\ \Phi(W_0 + \sum W_i X_i) &\leq C \quad \text{or} \quad > C, \\ W_0 + \sum W_i X_i &\leq K_0 \quad \text{or} \quad > K_0. \end{aligned}$$